

Reinforcement Learning – Imitation Learning – Assignment 1

Assignment Goal

The main goal of this assignment is to deepen your understanding of **Imitation Learning**, which is one of the central topics in the first part of the course.

To do so, you will first train an expert RL agent to solve the **LunarLander environment**. You will then use this expert to collect demonstration data and train a supervised model that learns to imitate the expert's policy. Unlike Reinforcement Learning, where learning is based on trial and error and reward signals, Imitation Learning relies directly on examples of state-action pairs generated by an expert.

A secondary goal is to give you some initial hands-on experience with Gymnasium environments and standard RL libraries, and to help you become familiar with the basic workflow of training and evaluating agents in reinforcement learning. The assignment is therefore designed both to strengthen your understanding of **Imitation Learning** and to expose you to the practical tools commonly used in RL experimentation.

Environment Description

The environment used in this assignment is:

LunarLander-v3

https://gymnasium.farama.org/environments/box2d/lunar_lander/

This environment simulates a spacecraft that must be landed safely between the flags while maintaining stability, controlling its speed, and managing fuel usage.

Part A — Training an Expert RL Agent

In this part, your goal is not to study the details of Reinforcement Learning algorithms, but rather to gain practical familiarity with the RL setting and the LunarLander environment. You should use a standard Reinforcement Learning library, such as Stable-Baselines3, to train an expert agent for the LunarLander-v3 environment.

At this stage, you are required to:

1. Choose a standard RL library and train an agent for LunarLander.
2. Verify that the agent improves during training and reaches reasonable performance.
3. Plot training results, including reward as a function of training progress.
4. Define clearly what you consider to be good expert performance.
For example: an average reward of 200 or higher over several evaluation episodes.

The purpose of this stage is mainly to let you experience the RL workflow:

- interacting with the environment,
- training an agent,
- observing reward curves,
- and evaluating the learned policy.

This stage is intentionally introductory and is not the main focus of the assignment. The main focus begins in the imitation learning stages that follow.

You may choose any standard RL library, modify hyperparameters, etc.

You (still...) are not required to:

- Implement an RL algorithm from scratch, understand the mathematical details of the RL algorithm,

Your notebook and report in this part should include:

- a short description of the RL library and setup you used,
- a graph showing training progress,
- a brief explanation of how you evaluated the expert agent,
- the performance criterion you selected.

Part B — Imitation Learning

After training an expert RL agent, you will use it to generate demonstration data and then train a supervised learning model that imitates the expert's behavior.

The main idea is simple: instead of learning through rewards and direct interaction with the environment, the imitation model learns from examples of **state–action pairs** generated by the expert agent.

The goal of this part is to examine how well a neural network can approximate the expert policy, and how its performance depends on the amount and quality of the demonstration data.

B1. Collecting Demonstration Data

Run the trained expert agent in the **LunarLander-v3** environment and collect pairs of **(state, action)**.

You should save the collected data and construct several datasets of different sizes.

At this stage, you are required to:

1. Run the trained expert agent for multiple episodes.
2. Record the state observed at each step and the action selected by the expert.
3. Build several datasets with different numbers of examples.
4. Describe clearly how the data was collected.

Your notebook and report should include:

- a description of the data collection process,
- the number of episodes used,
- the number of samples in each dataset,
- any decisions you made regarding dataset construction.

B2. Training an Imitation Model

Using the collected data, build a neural network that predicts the expert's action from the current state.

This stage should be treated as a standard supervised learning problem.

At this stage, you are required to:

1. Train a neural network to predict the action from the state.
2. Monitor the training process using appropriate training metrics.
3. Compare results across different dataset sizes.

You may choose the model architecture and training settings freely.

Your notebook and report should include:

- a description of the model architecture, the training setup,
- training and validation curves,
- a short explanation of the design choices you made.

B3. Evaluating the Imitation Model

After training the imitation model, run it as a policy in the **LunarLander-v3** environment. The goal of this stage is to evaluate how well the learned model can actually control the spacecraft, and not only how well it performs on the offline dataset.

At this stage, you are required to:

1. Load the trained imitation model and run it in the environment for multiple evaluation episodes.
2. Measure its performance and compare its behavior and results to those of the expert agent.

Your evaluation should include:

- average reward, number of steps until the episode ends, additional metrics if relevant.

B4. Effect of Dataset Size

Train several imitation models using different dataset sizes, and analyze how the amount of demonstration data affects performance.

The purpose of this stage is to understand the relationship between data availability and policy quality.

You should present graphs such as:

- dataset size vs. average reward,
- dataset size vs. success rate,
- dataset size vs. validation performance.

You may add any additional plots that help illustrate the differences between the models more clearly.

B5. Challenge & competition Section — Few-Shot Imitation Learning

This is the competitive part of the assignment.

Try to achieve the best possible imitation performance while using the **smallest possible number of training examples**.

In this part, you should:

- explore how to select especially useful examples,
- compare random selection with more deliberate selection strategies,
- present performance as an average over multiple runs, rather than based on a single run.

In your report, discuss:

- how the examples were selected,
- whether the selection was random or intentional,
- what worked well,
- what did not work well.

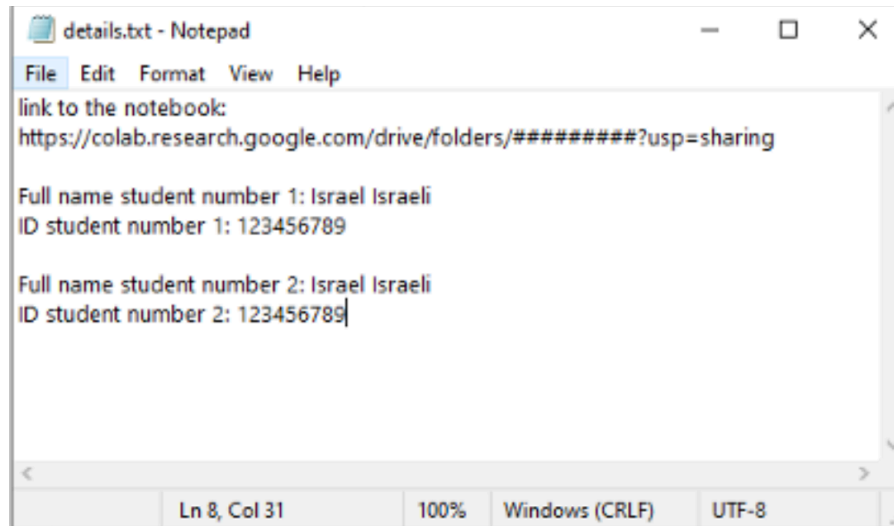
You must also answer the following question explicitly:

What matters more in this task: data quantity or data quality?

Submission Guidelines

1. Add video clips (In the notebook) that show relevant progress after the learning completed and converged.
 2. Even if you did not fully solve an environment - supply graphs\tables describing the average rewards on 100 episodes (after finishing the training phase).
 3. Submit a final report in pdf format when your ID numbers are included in the file name.
 4. You need to supply your code as Google Colab notebooks link. Such that the course team can run it.
 5. Write clean code! Separate code cells and make extensive use of text cells. A sloppy notebook will result in a lower grade.
 6. Do not change anything in notebook after the final submission date. A notebook that has been run/changed after the submission date will be automatically disqualified.
 7. The final report must be a professional, self-contained scientific document that fully explains your methodology and results without the need to reference the code.
- The report will be submitted to the submission box. The file name of the report will contain your IDs as follows: report_ID1_ID2.pdf
 - Do not forget to include the project title, your name and ID in this file.
 - Max number of pages: 8 (but you don't have to use them all !!).
 - The notebook will be presented in an orderly and clean manner; it will contain separate code cells and text cells that explain the actions performed.
 - **** Very important **** - when submitting, the notebooks will contain all the outputs relevant to the training results.
 - To the submission box you may also submit an explainer file that contains instructions and explanations on how to operate your notebook and other relevant details that need to be known to those who want to use your notebook.

Enter your details in the text file named submit.txt, the address of your notebook, the names and ID of the two partners, as follows:



```
link to the notebook:
https://colab.research.google.com/drive/folders/#####?usp=sharing

Full name student number 1: Israel Israeli
ID student number 1: 123456789

Full name student number 2: Israel Israeli
ID student number 2: 123456789
```

To sum up, in the submission box, submit the following files:

1. details.txt
2. report_ID1_ID2.pdf
3. explainer.txt (optional)

Team size

The project will be performed in groups of 2 students

Submission deadline:

May-20-2026